Proficient in Inventor (80 Hrs)

S.No	Session	Topics	Type	Duration (min)
1	Session 1	**Introduction:** Understanding parametric design, Understanding feature based concept, Basic modeling concepts, Understanding History-Based Modeling and Dependencies, Adaptivity, Orthographic Projections, File Extensions, Inventor Workflow	Theory/Demo	2.0 Hrs
		GUI		
		Mouse Control		
		Functional Keys		
		View Control: View Cube, Navigation Wheel, Pan, Zoom, Rotate, Look At, Previous View, Next View,Projections, Visual Styles		
		Clean Screen		
		Switch		
		Tile Window		
		Cascade		
		New		
		Work Environment: Project Files, Application Options, Document Settings, File Open Options		
2	Session 2	Exercises included for practice	Practicals	2.0 Hrs
3	Session 3	Sketch Workflow	Theory/Demo	2.0 Hrs
		Creating 2D Entities: Line, Circle, Ellipse, Arc, Rectangle, Polygon, Fillet, Chamfer		
		Advanced Sketching: Spline, Equation Curve, Slot, Sketch Point, Centre Point, Text & Geometry Text, Creating 3D Entities, Sketch Dimensioning, Sketch Constraints		
4	Session 4	Exercises included for practice	Practicals	2.0 Hrs
5	Session 5	**Modifying Sketch:** Move, Copy, Rotate, Trim, Extend, Split, Scale, Stretch, Offset, Editing Spline	Theory/Demo	2.0 Hrs
		Sketch Pattern: Rectangular, Circular, Mirror		
		Projection		
		2D Projections: Project Geometry, Project Cut Edges, Project Flat Pattern, Project to 3D Sketch		
		3D Projections		
		Sketch Properties		
6	Session 6	Exercises included for practice	Practicals	2.0 Hrs
7	Session 7&8	**Work Feature:** Work Plane, Work Axis, Work Point, UCS, in line work features	Theory/Demo	4.0 Hrs
		Part Modeling: Overview on Part Modeling, Task Performed in Part Modeling, Part Browser, Part Template		
		Basic Created Features: Extrude, Revolve, Sweep, Loft, Coil		
		Primitives		
8	Session 9&10	Exercises included for practice	Practicals	4.0 Hrs
9	Session 11&12	**Placed Features:** Hole, Thread, Fillet, Chamfer, Shell, Draft	Theory/Demo	4.0 Hrs
		Advanced Created Features: Derive, Emboss, Rib, Decal		
		Solid Editing: Direct Edit, Combine, Split, Thicken/Offset		
		Pattern: Rectangular, Circular, Mirror		
10	Session 13&14	Exercises included for practice	Practicals	4.0 Hrs
11	Session 15&16	**Measure:** Distance, Angle, Loop, Area, Region Properties	Theory/Demo	4.0 Hrs
		Parameters		
		i-Logic		
		Place Feature		
		Create i-Part		
		Insert i-Feature		
		Insert Object		
12	Session 17&18	Exercises included for practice	Practicals	4.0 Hrs
13	Session 19&20	Assembly Modeling Environment	Theory/Demo	4.0 Hrs
		Assembly Browser: Modeling View, Assembly View, Filter, User Defined Folder		
		Assembly Components: Placements and Constraints, Assembly Relationship Error & Recovery Tools		
		Flexible Components		
		Adaptive Components		
		Working with Assembly Components: Copy Assembly Components, Assembly Components in Patterns, Move & Rotation of Assembly Components, Deleting Assembly Components		
		Replace Components		
		Inspection for Interference		

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14	Session 20&21	Exercises included for practice	Practicals	4.0 Hrs
15	Session 22&23	Advanced Assemblies Flexible Assemblies Level of Detail Representations (LODR) Simplified Assemblies Shrink wrap Assemblies	Theoy/Demo	4.0 Hrs
16	Session 24&25	Exercises included for practice	Practicals	4.0 Hrs
17	Session 26&27	Sheet Metal Defaults **Sheet Metal Features:** Face, Flange, Contour Flange, Lofted Flange, Contour Roll, Hem, Bend, Fold, Cut, Corner Seam, Rip, Unfold, Refold, Hole, Corner Round, Corner Chamfer, Copy Object, Punch Tool Create Flat Pattern, Convert to Sheet Metal	Theory/Demo	4.0 Hrs
18	Session 28&29	Exercises included for practice	Practicals	4.0 Hrs
19	Session 30	**Surface Editing Tool:** Stitch, Patch, Sculpt, Trim, Extend, Replace Face, Repair Bodies	Theory/Demo	2.0 Hrs
20	Session 31	Exercises included for practice	Practicals	2.0 Hrs
21	Session 32&33	**Creating Fundamental Views:** Base View, Projected View, Auxiliary View, Section View, Detail View, Overlay View, Draft View Alignment, Creating Sketches in Drawing **Modifying Sketches:** Break View, Breakout View, Slice, Crop Creating New Sheet **Annotate:** Dimensioning, Arrange Dimension, Retrieve Dimension, Notes, Text, GD&T Symbols,Tables,Balloon	Theory/Demo	4.0 Hrs
22	Session 34&35	Exercises included for practice	Practicals	4.0 Hrs
23	Session 36	Freeform Modeling	Theory/Demo	2.0 Hrs
24	Session 37	Exercises included for practice	Practicals	2.0 Hrs
25	Session 38	Inventor Studio	Theory/Demo	2.0 Hrs
26	Session 39	Exercises included for practice	Practicals	2.0 Hrs
27	Session 40	Assignment for evaluation	Practicals	2.0 Hrs